

Teacher Guide — Chapter 2 – Shaping of the Earth's Surface

Geography | Class 9

Period 1

Entry ticket

Name the three main layers of the Earth.

Teacher script

Good morning, class! Today we begin exploring how the Earth's surface is constantly transformed by powerful forces acting from within and on the surface. One of the most important ideas explaining these changes is the theory of plate tectonics, given by W.J. Morgan. This theory describes how large pieces of the Earth's crust move slowly over the molten mantle. Let us examine the Earth's layers: the crust, mantle, and core. The crust and the upper part of the mantle together form the rigid outer layer called the lithosphere. Beneath the lithosphere lies the asthenosphere, a hot, mobile layer of partially molten rock that allows the plates to move. The movement of tectonic plates is driven by convection currents in the mantle, where heat from the core causes molten material to rise and cooler material to sink. When these plates meet at plate boundaries, they interact in three ways: convergent boundaries where plates move towards each other (forming features like fold mountains such as the Himalaya), divergent boundaries where plates move away from each other (creating features like the Mid-Atlantic Ridge), and transform boundaries where plates slide past each other (such as along the San Andreas Fault).

Blackboard notes

Key Terms & Concepts:

- Theory of Plate Tectonics: Given by W.J. Morgan; describes movement of Earth's crust over the mantle.
- Landform: A natural feature on the Earth's surface formed by weathering, erosion, deposition, and crust movement.
- Earth's Layers: Crust, Mantle, Core.
- Lithosphere: Crust + upper part of the mantle.
- Asthenosphere: Hot, mobile layer of partially molten rock beneath the lithosphere.
- Plate Boundaries:
 1. Convergent (e.g., Himalaya form via continental collision)
 2. Divergent (e.g., Mid-Atlantic Ridge)
 3. Transform (e.g., San Andreas Fault)

Activity: Plate Boundary Discussion and Mapping (discussion)

In pairs, examine the world map showing major tectonic plates. Pick two plates, identify whether they carry continents or oceans, and discuss how convection currents in the mantle push and pull these massive slabs of solid rock a few centimeters per year.

Checkpoints

- Who formulated the theory of plate tectonics?
- What is the difference between the lithosphere and the asthenosphere?
- Name the three main types of plate boundaries and give an example for each.

Exit ticket

Define what a landform is in one sentence.

Homework

Write a short paragraph explaining how convection currents in the mantle cause tectonic plates to move.

Mentor moment

Remember that the Earth beneath our feet, though it feels completely solid and unmoving, is part of a grand, dynamic system that has been shifting and shaping our world for millions of years. Every mountain and valley tells the story of these slow, powerful forces!

Period 2

Entry ticket

Write down one difference you think exists between how mountains are formed by internal forces versus how rocks on the surface are broken down.

Teacher script

Welcome back, class! In our previous session, we explored internal forces like plate tectonics that build up major physical features from deep within the Earth. Today, we turn our attention outward to examine the external forces that continuously break down and reshape the Earth's surface: weathering and erosion. A landform is a natural feature on the Earth's surface formed by processes such as weathering, erosion, deposition, and the movement of the Earth's crust. Weathering is the process through which rocks on the Earth's surface break down into smaller pieces due to various processes. Crucially, weathering does not involve the movement of the broken material—it only involves breaking down rocks in place. Once those pieces are broken down, erosion takes over. Erosion is the process by which soil, rocks, and other surface materials are worn away and carried from one place to another by natural agents like water, wind, ice, or waves. Let us be very careful to avoid the common misconception that weathering involves the transport of material; movement is the defining feature of erosion!

Blackboard notes

External Forces Shaping the Earth:

1. Weathering: The process through which rocks on the Earth's surface break down into smaller pieces.

- Key Fact: Does NOT involve movement of the broken material.

2. Erosion: The process by which soil, rocks, and other surface materials are worn away and carried from one place to another.

- Natural Agents: Water, wind, ice, or waves.

3. Landform Definition: A natural feature on the Earth's surface formed by processes such as weathering, erosion, deposition, and crust movement.

Activity: Weathering vs. Erosion Sorting Discussion (discussion)

Divide students into small groups. Distribute scenario cards describing different geological events (e.g., rainwater cracking a rock in place versus a river carrying sediment downstream). Have groups classify each scenario as either 'Weathering' or 'Erosion' and explain why based on whether movement of material occurred.

Checkpoints

- What is the primary difference between weathering and erosion regarding rock material?
- Does weathering involve the movement of broken rock material across the Earth's surface?
- What are some natural agents responsible for erosion?

Exit ticket

Define weathering in your own words and state one key factor that distinguishes it from erosion.

Homework

Write a short paragraph explaining how weathering and erosion work together to shape landforms, ensuring you correct the misconception that weathering moves rocks from place to place.

Mentor moment

Geography allows us to read the history of our planet written right in the rocks and soil beneath our feet. Every time you notice a cracked stone or a river carrying silt, you are witnessing the patient, powerful forces of nature at work. Keep looking closely at the world around you!

Additional activities

Asthenosphere and Lithosphere Convection Demonstration (demo)

Fill the container with water and place it on the heat source. Add a small amount of sawdust or tiny plastic pieces to the water. Heat the bottom and observe how the warm water rises and cooler water sinks, creating convection currents. Explain that this mirrors how heat from the Earth's core moves the semi-molten asthenosphere and pushes the rigid lithosphere plates above.

Tectonic Plate Boundary Role-Play (role_play)

Divide the classroom into small groups. Assign each group a plate boundary type: Convergent, Divergent, or Transform. Have students physically move their hands or bodies to demonstrate how plates move relative to each other. For convergent, they push palms together to form fold mountains like the Himalaya. For divergent, they pull hands apart to create mid-ocean ridges. For transform, they slide hands past each other like the San Andreas Fault.

Weathering vs. Erosion Sorting Challenge (discussion)

Distribute index cards describing different surface processes (e.g., 'Rainwater dissolving rock minerals in place', 'Wind carrying sand grains across a desert', 'Ice cracking rocks in a mountain crevice'). Have students work in pairs to sort the cards into two categories: Weathering or Erosion, keeping in mind that weathering does not involve the movement of broken material while erosion does.

Landform Mapping and Hazard Identification (other)

Provide students with a blank or outline world map. Have them use markers to locate and mark major tectonic features discussed in the unit, such as the Himalaya (convergent boundary), the Mid-Atlantic Ridge (divergent boundary), and the San Andreas Fault (transform boundary). Then, have them shade regions prone to earthquakes and volcanoes based on plate boundaries.

Learning gaps

Weathering involves the movement of broken rock material across the Earth's surface. [high]

Diagnostic: A rock high on a mountain cracks due to freezing water, and the pieces stay right where they fell. Is this an example of weathering, erosion, or both?

- Use a side-by-side comparison table contrasting weathering (in-place breakdown) with erosion (transportation).
- Provide a concrete analogy such as crumbling a cookie on a plate versus sliding the cookie crumbs across the table with a finger.

Tectonic plates float on a completely liquid ocean of magma. [medium]

Diagnostic: What is the physical state of the asthenosphere beneath the tectonic plates?

- Clarify that the asthenosphere is 'semi-molten' or plastic-like, similar to thick toothpaste or warm caramel, rather than a free-flowing liquid.
- Use a diagram showing the rigid lithosphere floating on the ductile asthenosphere.

Landforms are permanent and never change once they are formed. [low]

Diagnostic: True or False: Once a fold mountain like the Himalaya is formed by colliding plates, it stops changing completely.

- Explain the continuous cycle where internal forces build up landforms and external forces (weathering and erosion) constantly wear them down.
- Show a worked example of how rivers erode mountains over millions of years.